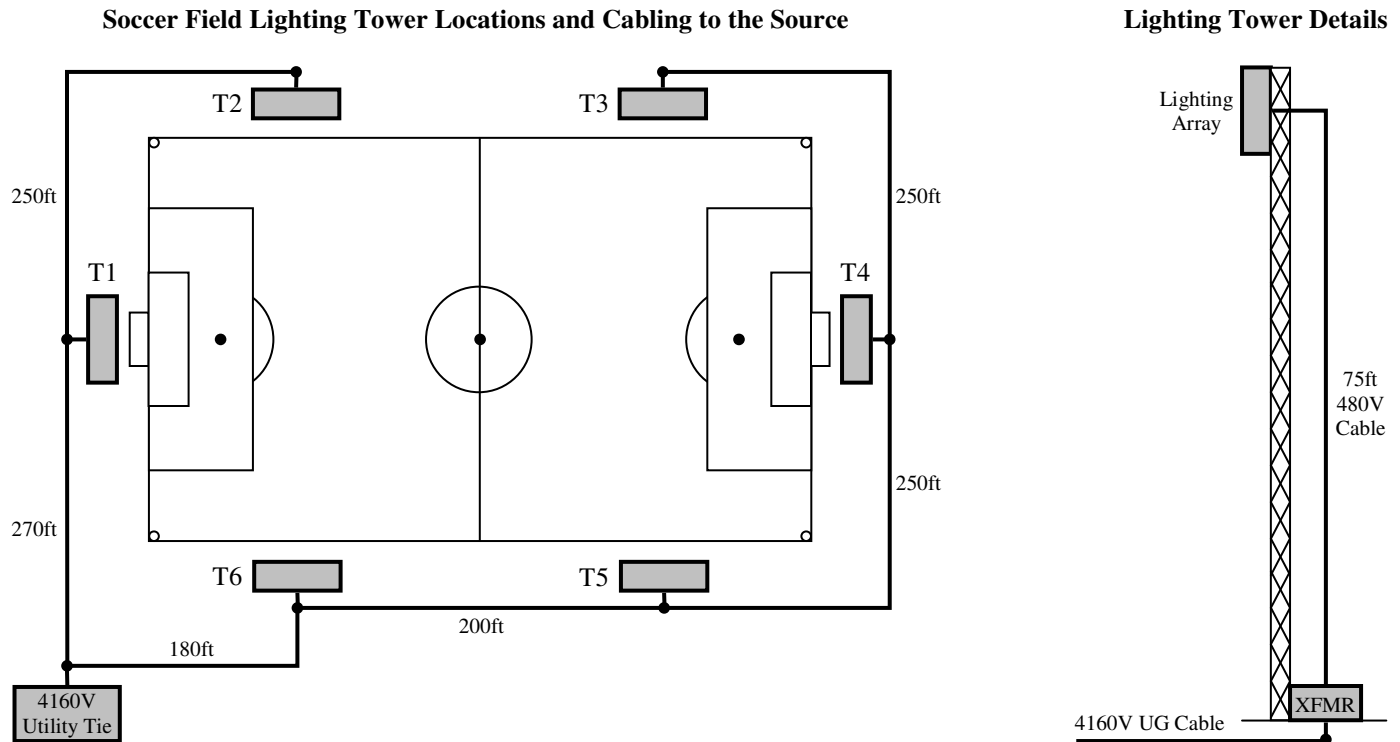


A 3 ϕ power system is being designed for the overhead lighting system of a soccer field. The site layout block diagram below shows six lighting array towers (T1, T2, T3, T4, T5, and T6) positioned around the soccer field. The lighting array towers are all powered from a 4160V electric utility company tie through a radial network of 3-conductor 4160V underground power cables as shown.



Notice the detail lighting tower diagram above and to the right. At the base of each lighting array tower, there is a Δ -Y connected 4160V/480V step-down transformer that lowers the voltage from a nominal 4160V level to a nominal 480V level. Then a 4-conductor 480V power cable is run from the transformer output terminals 75ft up the tower to the lighting array. The lighting array itself is a Y-connected load.

1. Draw a **special one line diagram** for the system that shows **only the source and the 4160V underground power cable network**. In this special one line diagram, **do not** show the details (XFMR, 75' cable, lighting array) of the lighting towers. Instead the lighting towers should be represented as simple loads labeled as "Load #T1", "Load #T2", etc.
2. Draw a **special one line diagram** for the system that shows **only one of the six lighting towers and its details (XFMR, 75' cable, lighting array)**. Model the 4160V underground cable feed in this special one diagram as a bus, and label it as "4160V UG Feed".
3. Draw a **wiring diagram** that corresponds to the one line diagram asked for in Part #2.

Each of the six lighting arrays is a 480V Y-connected load consisting of 54 individual 1500W, 277V sodium vapor lamps. The 54 lamps are split into 3 groups of 18 lamps. The 18 lamps in each group are wired in parallel. The 3 groups of parallel connected lamps are then connected in a 3 ϕ Y-configuration to create a 3 ϕ 480V Y-connected load with 277V (L-N equivalent of 480V) applied to each of the individual lamps.

4. Draw a **wiring diagram** for one of the lighting arrays. Use ditto (...) notation instead of drawing each of the 54 individual lamps.
5. Draw an **equivalent circuit** for the **entire power system**. Use the series impedance model for all of the power cables in the system.